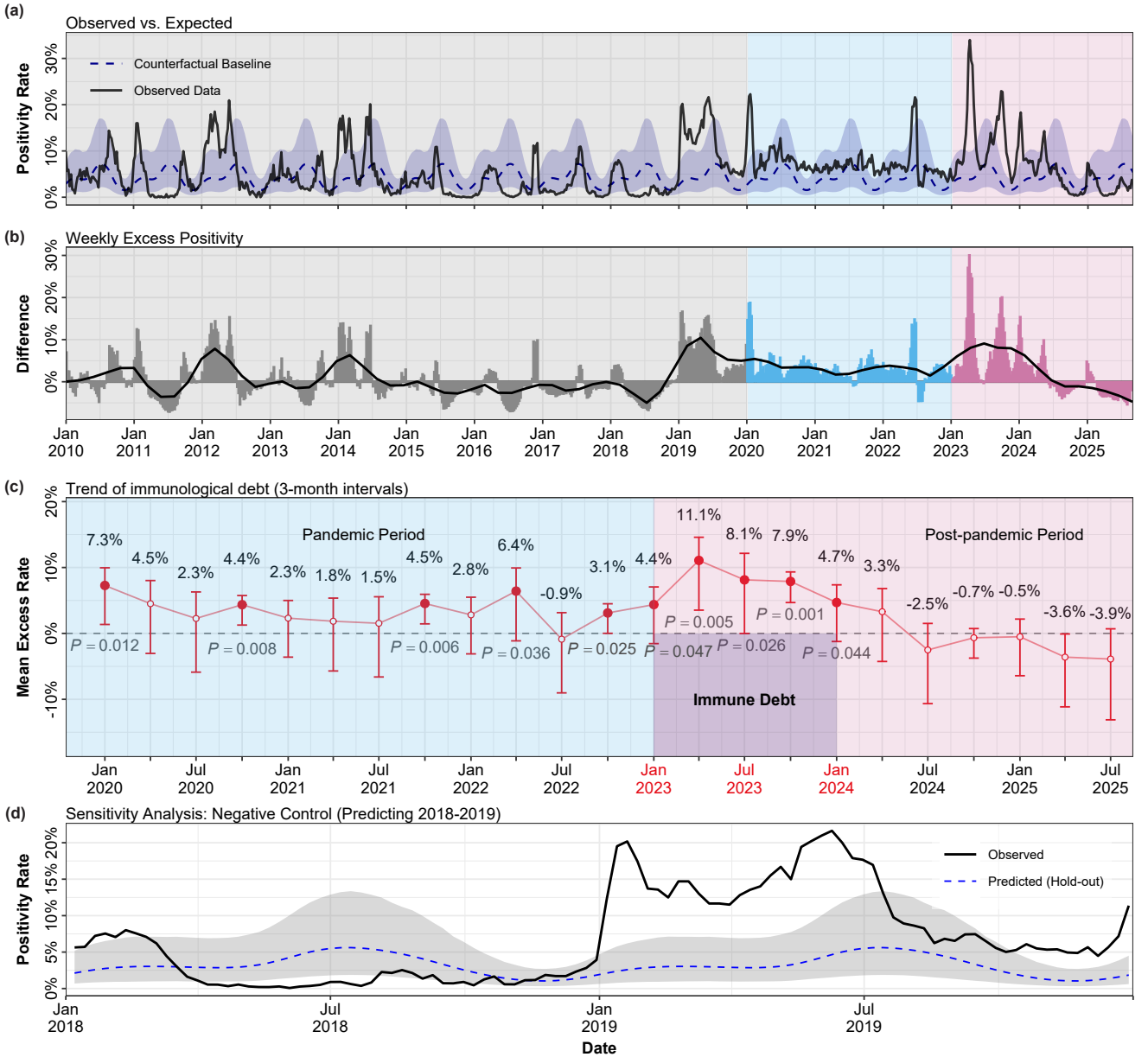

(a)
Observed vs. Expected
Positivity Rate
0%
10%
20%
30%
Counterfactual Baseline
Observed Data
(b)
Weekly Excess Positivity
Difference
0%
10%
20%
30%
Jan 2010
Jan 2011
Jan 2012
Jan 2013
Jan 2014
Jan 2015
Jan 2016
Jan 2017
Jan 2018
Jan 2019
Jan 2020
Jan 2021
Jan 2022
Jan 2023
Jan 2024
Jan 2025
(c)
Trend of immunological debt (3-month intervals)
Mean Excess Rate
20%
10%
0%
-10%
Pandemic Period
Post-pandemic Period
Immune Debt
7.3%
4.5%
2.3%
4.4%
2.3%
1.8%
1.5%
4.5%
2.8%
6.4%
-0.9%
3.1%
4.4%
11.1%
8.1%
7.9%
4.7%
3.3%
-2.5%
-0.7%
-0.5%
-3.6%
-3.9%
P = 0.012
P = 0.008
P = 0.006
P = 0.036
P = 0.025
P = 0.047
P = 0.005
P = 0.026
P = 0.001
P = 0.044
Jan 2020
Jul 2020
Jan 2021
Jul 2021
Jan 2022
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Jan 2025
Jul 2025
(d)
Sensitivity Analysis: Negative Control (Predicting 2018-2019)
Positivity Rate
0%
5%
10%
15%
20%
Observed
Predicted (Hold-out)
Jan 2018
Jul 2018
Jan 2019
Jul 2019
Date